

FINAL EXAMINATION

Paper: COMP 201 – Computer Programming using Python
Programme: ADA ADS (Mathematics) Session: 2021-2023 Semester: III

Maximum Marks: 60

Time Allowed: 150 Minutes

Note:

- Write, before solving the paper, your roll number with signature here: _____
- DO NOT write anything except the roll number and signature on the question paper.
- Write the syntax/wording of the code neatly and cleanly.
- Each mark and symbol is meaning full in a source code. So use them carefully.
- Use indentations while writing computer programs, appropriately.

Q.1	(a) What is different between following 2 statements: <pre>from directsolve import dolittle , crout from directsolve import *</pre> (b) What is output of following snippet: <pre>L1 = [3, 5, 9, 10, 15, 3.9, 1] L1[4] = L1[4]*9 L1[-2] = L1[-2]*10 print(L1)</pre> (c) Write Python expression for the following: $\frac{4y^9 - 8 \ln x^2 + \ln^2 y}{\sqrt{x} + 3}$ (d) Differentiate between count-controlled and sentinel-controlled repetition structures. Write a Python statement (or general structure) for each of these two. (e) Write a simple snippet that creates a one-dimensional NumPy array of 6 elements and then add a scalar value to each element of the array using broadcasting. <table border="1"><tr><td data-bbox="247 1368 542 1816">(f) What is output of following snippet: # easy question y = 5 x = y print(id(y)) print(id(x)) x = y**3 print(id(x)) print(id(y))</td><td data-bbox="542 1368 997 1816">(g) What is output of following snippet: # This question is hard # This question is tricky # Lets see how you do # Lets see how you work L1 = [3, 5, 9, 1, 15, 3.9, 1] L1[3] = L1[3]*9 L1[-3] = L1[-3]*10 print("Listtttttt=", L1)</td><td data-bbox="997 1368 1340 1816">(h) What is output of following snippet: a = [6, 7, "c", 9.8] b = ["pen", 4, 8, 1.1, 0] C = a + b C1 = sum((a , b) , []) print(C) print(C1)</td></tr></table> (i) Explain the difference between a void function and a value-returning function. Also tell how the code structures of these two types of functions differs from each other. (j) What is the output of the following snippet: <pre>Doc = [9, 2, 11, 3, -6, 0, 7, 4] ; print(Doc[6:2:-2])</pre>	(f) What is output of following snippet: # easy question y = 5 x = y print(id(y)) print(id(x)) x = y**3 print(id(x)) print(id(y))	(g) What is output of following snippet: # This question is hard # This question is tricky # Lets see how you do # Lets see how you work L1 = [3, 5, 9, 1, 15, 3.9, 1] L1[3] = L1[3]*9 L1[-3] = L1[-3]*10 print("Listtttttt=", L1)	(h) What is output of following snippet: a = [6, 7, "c", 9.8] b = ["pen", 4, 8, 1.1, 0] C = a + b C1 = sum((a , b) , []) print(C) print(C1)	20 = 2 × 10
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Q.2	Write a Python program that asks and receives a measure of the radius r of a circle and then computes and prints the area (πr^2) and circumference ($2\pi r$) of the circle.	8
Q.3	Write a program that asks the user to enter the value of marks (out of 100) and then computes and prints grade point (GP), according to the following formula: $GP = \begin{cases} 0.0, & \text{if marks} < 50 \\ 2.0 + \frac{(\text{marks} - 50)}{1.5} \times 0.1, & \text{if } 50 \leq \text{marks} < 80 \\ 4.0, & \text{if marks} > 80 \end{cases}$ Note that the input marks are rounded up, by taking ceiling of the marks. For example, each of 79.1, 79.5, and 79.9 would be rounded up to 80.	8
Q.4	Functions in Python can be categorized into four categories from a programmer's perspective. Write briefly about those 4 categories.	4
Q.5	What is the output of the following snippet? <pre>List = [[4 , 2 ,1 ,6] , [5 , 9 , -3] , []] L1 = List[0 : 2] L2 = List[: 1] L3 = List[1 :] print(L1) print(L2) print(L3)</pre>	3
Q.6	Write a program to ask the user to enter the charges for the CAR WASH, and then it displays 100 rupees as tip, amount of 6 percent sales tax, and amount of total charges.	4
Q.7	Write a program that asks and receives the length and width of a set of n rectangles and then compute and print the areas of the rectangles. Write a Python Function to obtain the area of the rectangle.	6
Q.8	An electric power supply company charges Rs. 7.74 per unit (KW/Hour) for the first 100 units of electricity consumed, Rs. 10.06 per unit for the next 100 units (i.e., units from 101 to 200), and Rs. 12.15 per unit for the rest of the units (i.e., units after 200) from the consumers. Write a program that receives the number of units consumed for n number of consumers and finds the electricity charges for each consumer, using for loop.	7

FINAL EXAMINATION (SAMPLE)

Paper: COMP 201 – Computer Programming using Python

Programme: BS (Mathematics) at Colleges

Session: Sample

Semester: III

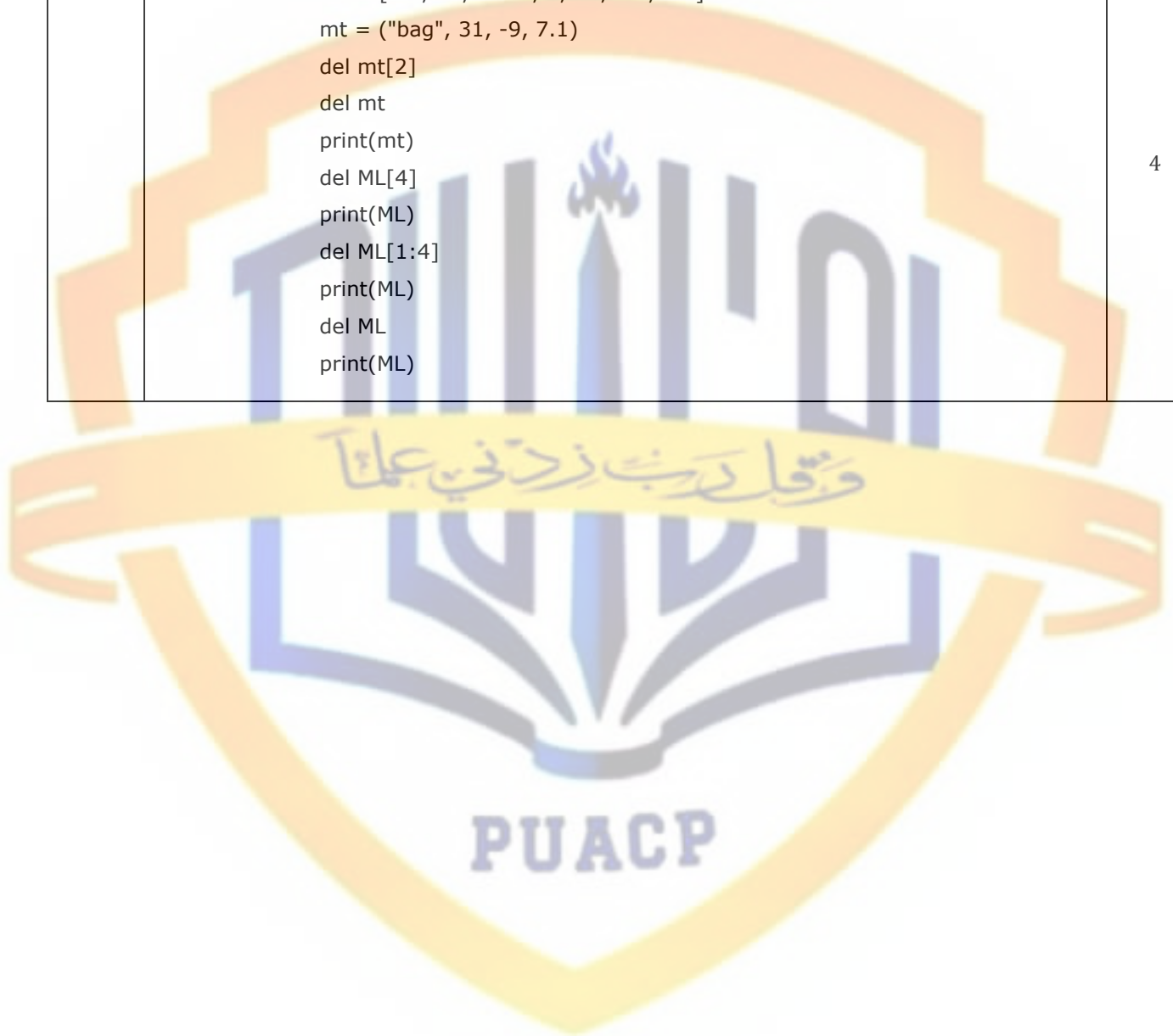
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Time Allowed: 150 Minutes

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Q.1	Write a program that asks the user for the number of passed out students and the number of failed students in a class. The program then displays the percentages of passed-out and failed students.	6																																	
Q.2	Write a note on the following related to Python: Script, module, package, library, framework	5																																	
Q.3	Write a Python script to compute the tax chargeable on the capital gain on selling an immovable property. The tax rate on capital gains on sale of an immovable property is given below: <table><tr><th>S. No.</th><th>Amount of gain</th><th>Rate</th></tr><tr><td>1</td><td>Where the gain does not exceed Rs. 5 million</td><td>3.5%</td></tr><tr><td>2</td><td>Where the gain exceeds Rs. 5 million but does not exceed Rs. 10 million</td><td>7.5%</td></tr><tr><td>3</td><td>Where the gain exceeds Rs. 10 million but does not exceed Rs. 15 million</td><td>10%</td></tr><tr><td>4</td><td>Where the gain exceeds Rs. 15 million</td><td>15%</td></tr></table> Moreover, the amount of any gain arising on disposal of an immovable property is computed, for tax purposes, according to the following formula: <table><tr><th>S. No.</th><th>Holding Period of Property</th><th>Taxable Gain</th></tr><tr><td>1</td><td>Where the holding period of an immovable property does not exceed one year</td><td>100%</td></tr><tr><td>2</td><td>Where the holding period of an immovable property exceeds one year but does not exceed two years</td><td>75%</td></tr><tr><td>3</td><td>Where the holding period of an immovable property exceeds two years but does not exceed three years</td><td>50%</td></tr><tr><td>4</td><td>Where the holding period of an immovable property exceeds three years but does not exceed four years</td><td>25%</td></tr><tr><td>5</td><td>Where the holding period of an immovable property exceeds four years</td><td>0%</td></tr></table>	S. No.	Amount of gain	Rate	1	Where the gain does not exceed Rs. 5 million	3.5%	2	Where the gain exceeds Rs. 5 million but does not exceed Rs. 10 million	7.5%	3	Where the gain exceeds Rs. 10 million but does not exceed Rs. 15 million	10%	4	Where the gain exceeds Rs. 15 million	15%	S. No.	Holding Period of Property	Taxable Gain	1	Where the holding period of an immovable property does not exceed one year	100%	2	Where the holding period of an immovable property exceeds one year but does not exceed two years	75%	3	Where the holding period of an immovable property exceeds two years but does not exceed three years	50%	4	Where the holding period of an immovable property exceeds three years but does not exceed four years	25%	5	Where the holding period of an immovable property exceeds four years	0%	10
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Q.4	Explain the difference between break and continue statements.	4																																	
Q.5	A 160-pound person burns approximately 15.1 calories per minute running. Use a loop to display number of calories burnt after 10, 15, 20, and 25 minutes.	6																																	
Q.6	Write down 4 ways of Joining or Concatenating Two or More Lists/Tuples with example codes.	8																																	
	Write a Python program that receives the input value of a function f defined by $f(x) = \begin{cases} 2x^2 + 3x + 4, & \text{for } x < 2 \\ 0, & \text{for } x = 2 \\ 2x^2 + 3x - 4, & \text{for } x > 2 \end{cases}$ Write a user-defined function $fx()$ that receives the value of x and returns the value of function f at x, i.e., $f(x)$.	6																																	
Q.7	An electric power supply company charges its residential consumers for the electricity units according to the tariff as below: Rs. 7.74 per unit (KW/Hour) for the first 100 units,	7																																	

	Rs. 10.06 per unit (KW/Hour) for the next 100 units (i.e., units from 101 to 200), Rs. 12.15 per unit (KW/Hour) for the next 100 units (i.e., units from 201 to 300), Rs. 19.55 per unit (KW/Hour) for the next 400 units (i.e., units from 301 to 700), and Rs. 22.65 per unit for rest of the units (i.e., units after 700). Write a program that asks the user to enter the number of units consumed and finds the electricity charges for an arbitrary number of consumers.	
Q.8	Explain the concept of list comprehension with mapping , and list comprehension with mapping	4
Q.9	What is the output of the following snippet: <pre> ML = [54, 88, -71.1, 0, 19, 4.9, 12] mt = ("bag", 31, -9, 7.1) del mt[2] del mt print(mt) del ML[4] print(ML) del ML[1:4] print(ML) del ML print(ML) </pre>	4



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Q.2	Write names of any 4 non-primitive data types. Also declare a variable of each of these data types and assign them an object/value/instance of that type.	8			

Q.3	Write a program that asks the user to enter the value of marks (out of 100) and then computes and prints grade point (GP), according to the following formula: $GP = \begin{cases} 0.0, & \text{if marks} < 50 \\ 2.0 + \frac{(\text{marks} - 50)}{1.5} \times 0.1, & \text{if } 50 \leq \text{marks} < 80 \\ 4.0, & \text{if marks} > 80 \end{cases}$ Note that the input marks are rounded up, by taking ceiling of the marks. For example, each of 79.1, 79.5, and 79.9 would be rounded up to 80.	8
Q.4	Functions in Python can be categorized into four categories from a programmer's perspective. Write briefly about those 4 categories.	4
Q.5	What is the output of the following snippet? <pre>gg = [] n = 8 for i in range(n): gg.append(i) print(gg)</pre> Using list comprehension, write a replacement of this code to get the same output.	2 + 2
Q.6	Write a program to ask the user to enter the charges for the food, and then it displays the amounts of a 10 percent tip, 5 percent sales tax, and total charges.	4
Q.7	A company pays its employees weekly according to their hourly wage up to 40 hours, and 50% more for the additional hours (overtime). Write a program that uses a while statement to receive values of "hourly_rate" and "hours_worked" for an arbitrary number of employees and finds and prints the gross pay for each employee.	6
Q.8	Write a Python code program to approximate the value of sine of an angle x by using the formula (input x in radians): $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} = \sum_{k=1}^n (-1)^{k-1} \frac{x^{2k-1}}{(2k-1)!}$	6